

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I am looking for a controller seal material

Assistant: To help identify the best material for your controller seal, could you please clarify:

1. What type of controller is this seal intended for? For example, is it for an industrial PLC, a consumer electronics device (like a game controller), an automotive controller, or another type?
2. Where will the controller be used? Is it in an indoor, outdoor, or harsh environment (such as exposure to dust, moisture, chemicals, or temperature extremes)?

This information will help ensure the material suggestion fits your specific application and environment.

User: i dont have this information

Assistant: Since you don't have details about the controller type or environment, I'll assume typical conditions for controller seals in electrical and electronics applications (indoor/outdoor, moderate exposure to dust and moisture, elastomeric material).

To proceed, could you please provide:

1. What is the minimum wall thickness of the seal you require? If you don't know, just let me know.

2. Do you have any preferred base material for the seal (such as silicone, EPDM, or other elastomers)?
3. Are you open to alternative material suggestions if your preferred material is not available?

User: i dont have prefrence but I need good bonding with nylon. And a Shore 50

Assistant: Here are the key requirements assumed for your controller seal application. Please confirm if these are correct for your use case:

1. Shore A hardness: 50 (as you specified)
2. Processing requirement: Injection molding
3. Feature: Good bonding to nylon (PA6/PA66)
4. Mechanical property: Compression set resistance
5. Material resistance: Chemical/fluid resistance
6. Durability: Heat stabilized or stable to heat
7. Environmental exposure: Humidity/water resistance (low water uptake)
8. Processing requirement: Ultrasonic weldable / 2K overmolding compatibility

Are all of these requirements correct for your application? Please let me know if anything needs to be changed or if you have additional requirements.

User: it should have low creape and high flow

Assistant: Thank you for confirming the requirements and adding low creep and high flow. To further refine the material selection, could you please clarify the following:

1. Is UL flammability or glow-wire certification required for this seal (for example, UL94 V-0 or IEC glow-wire)?
2. What is the minimum wall thickness of the seal? If you don't know, please let me

know so I can make an assumption.

3. Will the seal be exposed to sunlight or ozone (outdoor or harsh environment)?

Your answers will help ensure the material suggestion fits your needs. If you are unsure about any, just let me know.

User: its for indoor and dont know the thickness

Assistant: Thank you for clarifying that the seal is for indoor use and you don't know the minimum wall thickness. To finalize the requirements, could you please confirm:

1. Is UL flammability or glow-wire certification required for this seal (such as UL94 V-0 or IEC glow-wire)?
2. Are any electrical properties needed, like anti-static or static dissipative features?
3. Is hydrolysis resistance important for your application?

If you are unsure about any of these, just let me know.

User: these things are not req suggest grades

Assistant: Based on your requirements, here are two potential options for your controller seal:

1. SANTOPRENE® 8211-45 (Rank 1)

- Shore A hardness: 49 (very close to your target of 50)
- Designed for injection molding with high flow, ideal for thin and complex seal geometries
- Good compression set resistance and low creep
- Suitable for 2K overmolding onto nylon (PA6/PA66) with proper process optimization
- Excellent chemical and humidity resistance for indoor environments

2. SANTOPRENE® 111-45 (Rank 2)

- Shore A hardness: 49
- Injection molding grade, widely used for seals
- Good compression set and durability
- Can be overmolded onto nylon, though high flow is not explicitly listed

These grades may be suitable for your application based on technical data. Please ensure final material selection and product design decisions are validated for your specific use case. If your application involves critical safety or regulatory compliance, make sure to check all applicable standards and regulations, as Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.